

TIMESERIES CHANGE DETECTION OF MEGADROUGHT AND HUMAN EXTRACTION IN “EL YESO” RESERVOIR, CHILE

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ABSTRACT:

Megadroughts and human exploitation of water resources pose severe threats to freshwater resources globally. This paper addresses the critical need to understand and monitor the impacts of the ongoing megadrought on Chile's "El Yeso" reservoir, a key water source for Santiago and its agricultural and human consumption needs. While previous work by Fuentealba et al. (2025) documented significant surface area loss (16%) over the last 35 years, it lacks detailed analysis of year-by-year changes in water levels. By applying the Breaks For Additive Seasonal and Trend Lite (BFAST Lite) algorithm to long-timeseries Landsat (1985-2025) Normalized Difference Water Index (NDWI) composites, it is possible to detect spatial and temporal changes in the “El Yeso” reservoir surface area. Breakpoint analysis shows the timing of these changes aligns with the period of most rapid surface area decline observed by Fuentealba et al. (2025), particularly from 2015-2020, and also detected distinct breakpoints in between 2008 and 2010 corresponding to shifts in the long-term stable trend. The study demonstrates the algorithm's use for detecting changes in water bodies but highlights the necessity for integrated analysis incorporating climate, extraction, and hydrological data to establish causal links for effective water resource management in water crises.

1. INTRODUCTION

Megadroughts are characterized as extended periods of drought spanning decades and are being exacerbated by human-induced climate change (Ault et al., 2016). Since 2010, a megadrought in Chile has brought water scarcity to Northern and Central regions causing near collapse of natural forests, decrease in surface and ground water levels, decrease in agricultural practices, and indirect social unrest (Garreaud et al., 2017; Garreaud et al., 2025). Water resources are a critical resource in ecosystems and their biodiversity, as well as to meet human needs in activities such as agriculture, mining and consumption. As a result, climactic and human-induced stresses directly affect Chile's main source of freshwater: lakes and reservoirs. These are understudied, with lacking localized information on how and when droughts and excessive extraction affect the waterbody (Fuentealba et al., 2025). Critical to Santiago's agricultural and human-consumption needs, the “El Yeso” reservoir has seen shrinkage between 2010-2021 (Fuentealba et al., 2025). Fuentealba et al. (2025) determines the megadrought is drying most of the lakes in Northern and Southern Chile, however, the change is being determined by comparing 2 time periods. This analysis describes a broad trend over a long period of time and across many lakes, however, it is limited in understanding year-by-year changes where the direct impacts of climactic and human-induced stresses can be observed. The paper does provide yearly surface area recordings that serve as a comparison metric to the change detection made by BFAST Lite. Additionally, there is more knowledge is required in testing the use of the BFAST algorithm in hydrology (Mendes et al., 2022). This paper investigates the use of Breaks For Additive Seasonal and Trend Lite (BFAST Lite) algorithm in detecting changes in lake surface area caused by the megadrought and human-induced effects (Masiliūnas et al., 2021).

1.1 Research question

Can BFAST Lite detect spatial and temporal changes caused by climactic or anthropogenic stresses on “El Yeso” Reservoir, Chile, using long timeseries of Landsat data (1985-2025)?

2. MATERIALS AND METHODS

2.1 Study Area

The “El Yeso” reservoir is located 58km East (as the crow flies) of Santiago, the capital of Chile. It is located within the Maipo river basin in the Metropolitan region of Chile. The reservoir was created with the damming of the Yeso river in 1964 and is filled through icecap melting and the Echaurren glacier (Dionizis Rojo, 2018). The reservoir has lost 1.35 km² or 16% of its surface area between the periods 1985-2009 and 2009-2021 as a cause of a megadrought (Fuentealba et al., 2025).

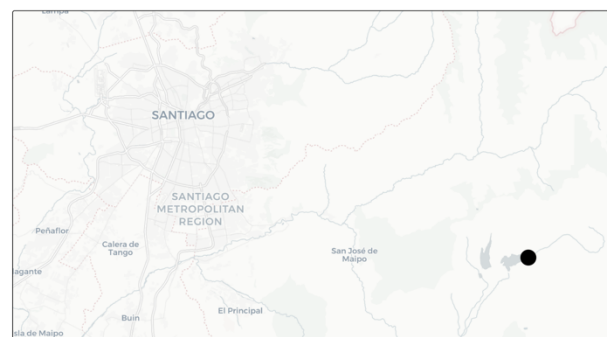


Figure 1. Location of area of study: “El Yeso” reservoir.

2.2 Data Acquisition & Preprocessing

The timeseries composites were acquired through Google Earth Engine (GEE) using Level 3 Normalized Differences Water Index (NDWI) 8-day composites (Google for Developers, n.a.). Composites from January to March were composited using the median as final value defining a yearly image for a timeframe from 1985 to 2025 (Appendix A). This choice of month timing was made to minimize cloud cover since it is the late-dry-season. The use of the Normalized Difference Water Index offers a reliable measurement of lake surface area because it detects open water bodies (McFeeters, 1996). It is defined as:

$$NDWI = \frac{GREEN - NIR}{GREEN + NIR} \quad (1)$$

As a result, preprocessing of cloud masking and NDWI calculation are attributed to Google's dataset (Google, n.a.). Finally, NDWI values lower than 0.15 were removed to focus the change detection on the reservoir area.

2.3 Change Detection

To be able to detect trend changes in the reservoir's area, the BFAST Lite model was implemented. It is a lightweight model based on the BFAST algorithm proposed by Verbesselt et al. (2009). BFAST was introduced as an unsupervised time series change detection algorithm, specialized in detecting multiple breakpoints in a timeseries (Verbesselt et al., 2009). This model separates trend and seasonality based on the following equation:

$$Y_t = T_t + S_t + e_t \quad (2)$$

where Y = observed data at time t
 T = trend component
 S = seasonal component
 e = remainder or noise component

However, the lightweight model is implemented to save time, since it skips the decomposition of trend and seasonality using locally estimated scatterplot smoothing (LOESS). BFAST Lite improves by being quicker and more flexible, divided in two steps: firstly, generating regressors for piecewise linear regression by fitting trend, harmonics, seasonality, and optional external features to the input data (with user-configurable parameters). Secondly, estimating breakpoints by creating multiple piecewise linear regressions and selecting the optimal number of breaks using the Bayesian Information Criterion (BIC) (Masiliūnas et al., 2021). Additionally, the regressors were set only to detect trends, since yearly data will not observe seasonality. The R code and composite data is provided in Appendix B. The model outputs the year and magnitude of the break, however, only magnitudes higher than 0.15 are presented to avoid breaks caused by noise in the NDWI timeseries.

3. RESULTS

The results present the year and magnitude of maximum change detected. Figures 2 and 3 show most of the breaks occurring after 2008 located at the borders of the lake with the highest density of breakpoints occurring at the most upstream region of the reservoir. Figure 3 shows the most recorded breaks occurring in 2014 and 2016 located upstream and on the southern flank of the reservoir. Moreover, a small peak in changes was recorded between 2008 and 2010. The breaks occurring before 2008 were primarily recorded in 1986, 1991-1993 and 1997 in the centre of the lake.

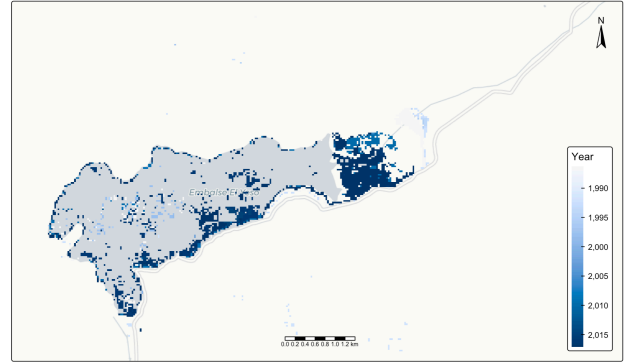


Figure 2. Year of Maximum Change Detected between 1985 and 2025 in the "El Yeso" Reservoir.

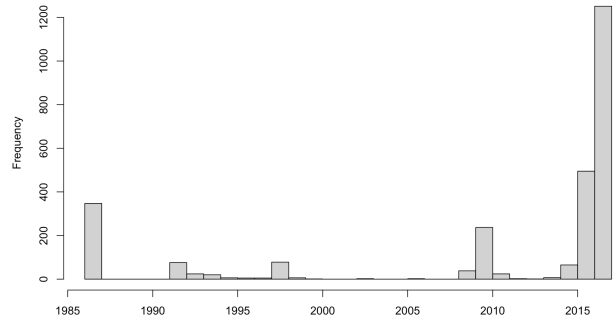


Figure 3. Frequency of Maximum Breakpoint Detected between 1985 and 2025 in the "El Yeso" Reservoir.

Figure 4 presents maximum magnitude of the breakpoints, showing the highest recorded breakpoints are located upstream, with a majority being higher than 0.5 in magnitude. Conversely, the lowest magnitude values recorded are observed inside the reservoir and away from the banks.

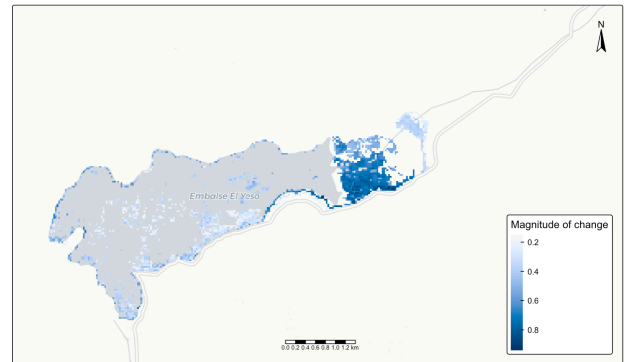


Figure 4. Maximum Magnitude Change Detected between 1985 and 2025 in the "El Yeso" Reservoir.

4. DISCUSSION

The results of the BFAST Lite break detection concludes on 3 key takeaways which will further be discussed: 1) significant portions of the reservoirs border contain change detection; 2) changes occur primarily at the opening of the Yeso river in between 2014 and 2016; and 3) high magnitude of change is mostly observed at the opening of the Yeso river.

Firstly, the figures conclude that BFAST Lite is reliable in detecting changes occurring to water surface levels, using 30m resolution Landsat data, masking out NDWI values lower than

0.15 and magnitude values lower than 0.15. These settings provided clear results, highlighting the success in detecting change in the “El Yeso” reservoir. However, there is a possibility of factors affecting different NDWI measurements, for example, measurement errors caused by different Landsat satellites, sensor noise, biochemical changes in the water such as mineral content. Conclusions made on the effect of the megadrought, or other human induced stresses of dam management and water extraction are observed but not clearly attributed to a particular event.

Secondly, most changes were observed from 2014 to 2016 and a small peak between 2008 and 2010. It is worth noting that the application of BFAST Lite is focused on a breakpoint moment in time where the trend changes, therefore, continuous years of decrease in lake surface area won't cause a breakpoint, because it becomes the new trend. Fuentealba et al. (2025) surface area calculations show a decrease in surface area from 2005 onwards. In comparison, the BFAST Lite detection of change was primarily concentrated in the 2008 and 2016 period. In addition, most decrease in surface area recorded by Fuentealba et al. (2025) was between 2015 and 2020 which matches the high frequency in detections between 2015 and 2016. To summarize, a smaller decreasing trend in surface area was recorded in 2005 and detected in 2008, and a larger decreasing trend is observed in 2015 and detected in 2015.

Thirdly, the highest magnitude changes were detected at the opening of the Yeso river and recorded between 2014 and 2016 that is also observed in Fuentealba et al. (2025), where the steepest decrease in surface area occurred from 2015 onwards. The location of highest change can be attributed to the topography of the reservoir, having steep walls on each side and a low relief topography at the river opening. Therefore, an increase or decrease in surface area will be more visible in low relief areas.

Even though the model can detect the mayor changes in surface area of the reservoir, it is difficult to differentiate between climactic or human dependent changes. Additional research must be developed using more datasets such as climactic and water extraction data to find possible correlations.

To improve BFAST Lite detection conclusions, further research can specify if the change detection refers to a positive or negative change in surface area. Additionally, limitations in the model can overlook gradual changes characteristic to droughts, therefore, deeming the tool unsuitable in detecting the effect of droughts on Earth. Models like LandTrendr can offer tailored results to trend detection time series (Kennedy et al., 2010).

5. CONCLUSION

The megadrought in Northern and Central Chile has caused significant environmental and socio-economic damage such as a near collapse of natural forests, decrease in surface and ground water levels, decrease in agricultural practices, and indirect social unrest (Garreaud et al., 2025). Therefore, it is important to monitor the effect of droughts and human water consumption on lake and reservoir surface area to correctly manage the natural resource supplying 8 million people with drinking water. This paper concludes that BFAST Lite is successful in detecting changes in water surface area, however, determining causal relationships to droughts or human extraction requires further research.

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5.1 Use of generative AI statement

Claude, Gemini models were used using cloud computing, and QWEN3 with 4billion parameters was used locally. These models were used as partners in determining the feasibility of a research idea, to help suggest code structures, and to summarize information. Code was created in Javascript (GEE) and R, first to suggest a basic download structure, second to help in setting the basics for making maps using tmap package. Logs are provided in Appendix C.

5.2 Appendix

5.2.1 APPENDIX A. Google Earth Engine Data Acquisition.

5.2.2 <https://code.earthengine.google.com/3e06d1e0ea0725b29d78298c5ed7ebcd>

5.2.3 APPENDIX B. Source Data and Scripts.

<https://drive.google.com/drive/folders/1F9yxhuQKlpsorirwEBv31agjBFWb4blQ?usp=sharing>

5.2.4 APPENDIX C. Claude Chat Log.

https://drive.google.com/drive/folders/1-uXpLhLxYqxjpXBuW4S_DBEmEiKTlafz?usp=sharing